Transmitter source model (transmitter)

Statements digest, version 0.13.0

Every definition, fact, assumption, result and requirement of the contract, with the contract's numbers; remarks, examples, technical steps and all proofs are left out. Each heading opens its statement in the contract.

Mathematics

1 Source specifications and source modes

Definition 1.1 (Source specification)**.** A *source specification* $\mathfrak{S}_{\mathrm{src}}$ consists of the following data. A finite non-empty alphabet $\mathcal{X}$ with a declared order. A finite-dimensional system A with a declared computational basis, $d_A \coloneqq \dim \mathcal{H}(A)$, and a declared *sector structure* on it: an ordered finite index set $\mathbb{T}$ and a decomposition $\mathcal{H}(A) = \bigoplus_{t\in\mathbb{T}} \mathcal{H}(A_t)$ into subspaces each spanned by vectors of the declared basis, $d_{A_t} \coloneqq \dim \mathcal{H}(A_t) \geq 1$, with projectors Π_t^A; the *one-sector structure*, $\mathbb{T}$ a single index and $A_t = A$, is the default. A POVM $\{M_x\}_{x\in\mathcal{X}}$ on A presented as a block list against that structure, $M_x = \bigoplus_t M_x^{(t)}$ with $M_x^{(t)} \in \mathcal{L}(A_t)$, each block Hermitian and positive semidefinite, and $\sum_{x\in\mathcal{X}} M_x^{(t)} = \mathbb{1}_{A_t}$ for every t with the sum taken in the declared order. The *specified signal ensemble* $\{p_x, \rho_x\}_{x\in\mathcal{X}}$ on the *transmitted system* A', with $p_x \geq 0$, $\sum_x p_x = 1$ and ρ_x a state on A'. With one sector the block list M_x is the matrix M_x, and every condition on its blocks is the same condition on that matrix.

How the specified ensemble is declared. In one of two ways. (a) *As matrices*, for a finite-dimensional A': by its *weighted members* $\sigma_x \in \mathrm{Pos}(A')$, with $p_x \coloneqq \operatorname{tr}\sigma_x$, $\sum_x p_x = 1$ in the declared order, and $\rho_x \coloneqq \sigma_x/p_x$ where $p_x > 0$. A declaration by the pair (p_x, ρ_x), $\rho_x \in S_=(A')$, is the declaration of the weighted member $\sigma_x = p_x\rho_x$ with the product formed exactly, and it requires $\rho_x \in S_=(A')$ of *every* letter, a letter with $p_x = 0$ included. Every condition of this module that reads an admitted ensemble reads it through its weighted members only. (b) *By a source model*, for a transmitted system A' whose Hilbert space is separable, possibly infinite dimensional: by a named definition of a submodule of this module that fixes the weights p_x and the states ρ_x as mathematical objects and not as matrices, together with the physical-source assumption which that submodule declares in the place of Assumption 5.1. A specification declared by a source model is presented with a non-empty source segment (Definition 2.2): what a consumer receives is never the model but the virtual ensemble, which is finite dimensional.

The two modes. The specification is declared in one of two modes. In *trusted-source mode* it additionally carries a marginal $\tau_A \in S_=(A)$, presented

as a block list $\tau_A = \bigoplus_t \tau_A^{(t)}$ with $\tau_A^{(t)} \in \mathrm{Pos}(A_t)$ and $\sum_t \operatorname{tr} \tau_A^{(t)} = 1$, and, at $d_A \geq 2$, a source-replacement certificate in the sense of Definition 3.3, written against the *virtual* ensemble $\{p_x, \rho_x^{\mathrm{v}}\}$ on the virtual transmitted system A'': the finite-dimensional ensemble the source segment starts from, whose channels produce the specified ensemble from it (Definition 2.2 (S1), (S3)); when the source segment is empty, $A'' = A'$ and $\rho_x^{\mathrm{v}} = \rho_x$. The certificate takes one of the three forms of Definition 3.3: a rank-one projector (the pure form), positive sector blocks (the block form), or an ensemble presentation, which declares only the weighted virtual ensemble, in blocks along a matched sector structure of A'', and defines the certificate from it. A sector structure with more than one sector is declared only in trusted-source mode and only together with a certificate in block form or an ensemble presentation. *With an ensemble presentation* the specification declares neither A, its basis and its sector structure, nor $\{M_x\}$, nor τ_A: they are constituents of the specification that (3.2) defines from the declared virtual ensemble, the conditions this definition places on them hold by Lemma 3.4 (i), and of them only τ_A is formed as numbers, that is, computed entry by entry from the declared blocks (Lemma 3.4 (iii)); $\{M_x\}$ and the certificate are defined and never computed. At $d_A = 1$ no certificate is demanded and $\tau_A = [1]$ is the only marginal, so that there the two modes impose the same condition. In *untrusted-source mode* the specification carries no certificate and no marginal, written $\tau_A = \mathsf{none}$; the specification with $d_A = 1$ and a one-letter $\mathcal{X}$ is its degenerate case, in which Alice prepares nothing that this module describes.

What the sector structure is not. It is a declaration about $\{M_x\}$ and τ_A, which are block diagonal by their presentation. It is no statement about any operator on A together with a further system, and no consumer may read it as one: that a joint state is block diagonal along $\mathbb{T}$ is either proved of the states a certificate reaches (Lemma 4.5) or licensed by a theorem of the consuming module that restricts its own feasible set, under that module's own preconditions, and is never inferred from the presentation of τ_A or $\{M_x\}$.

The number model. Every declared entry is a binary64 number, hence a dyadic rational, and is admissible exactly as declared. A source model (case (b)) declares no entries: its weights and states are mathematical objects, and what is decided of such a specification is decided on the entries it does declare, those of its virtual ensemble among them. Every condition above is decided *exactly* on the declared entries: a quantity that a condition compares — a sum in the declared order, a product, a partial trace — is *formed* in exact arithmetic on the declared entries as well as compared exactly, so that the identity decided is the identity written and rounding enters nowhere. That the operands are binary64 is not what makes a decision exact; a product of two binary64 numbers need not be one. Positive semidefiniteness is decided by an exact decision on the declared entries and

never by a floating-point eigenvalue.

2 Source links and the source segment

Definition 2.1 (Source link)**.** A *source link* Ξ is a typed link of a reduction chain (Definition "Reduction chain" (reductions/def:chain)) at the stage source: a tuple

$$\Xi = \Big(\mathsf{kind},\ \big(A^{\mathrm{in}}, \{\sigma_x^{\mathrm{in}}\}_{x\in\mathcal{X}}\big),\ \big(A^{\mathrm{out}}, \{\sigma_x^{\mathrm{out}}\}_{x\in\mathcal{X}}\big),\ \Gamma_\Xi,\ \mathcal{D}_\Xi,\ \mathfrak{s}_\Xi,\ \mathfrak{L}_\Xi,\ \mathsf{licence},\ \mathsf{status}\Big) \tag{2.1}$$

whose slots are the following.

(a) The kind. kind is a source-stage link kind. The vocabulary is the kinds this module declares, of which there is exactly one, source-map, together with the kinds a submodule of this module declares in its own interface. A submodule that declares a kind supplies its soundness claim, and a kind for which no claim of this repository proves the link's identity is declared and refused (Definition 2.2, "Which kinds are admitted"), as source-map is. An admitted kind has an entry in the soundness registry of that definition, and a kind without one is not admitted whatever a link declares.

(b) The two descriptors. The *source descriptor* $(A^{\mathrm{in}}, \{\sigma_x^{\mathrm{in}}\})$ is the transmitted system and signal ensemble the link starts from and the *target descriptor* $(A^{\mathrm{out}}, \{\sigma_x^{\mathrm{out}}\})$ the one it produces, both indexed by the same alphabet $\mathcal{X}$ in the same declared order, and both read through their weighted members $\sigma_x = p_x\rho_x$, as every ensemble of this module is (Definition 1.1): σ_x^{in} and σ_x^{out} are positive semidefinite with $\operatorname{tr}\sigma_x^{\mathrm{in}} = \operatorname{tr}\sigma_x^{\mathrm{out}} = p_x$, which is what carrying the weights untouched means.

(c) The channel witness, in one of two ways. Γ_Ξ is a channel from A^{in} to A^{out}, supplied either *as data* or *as a theorem constructor*.

(c1) *As data*: both systems are finite dimensional and the channel is declared, so that the identity $\Gamma_\Xi(\sigma_x^{\mathrm{in}}) = \sigma_x^{\mathrm{out}}$ for every $x \in \mathcal{X}$ is an exact matrix identity on declared numbers and is decided as such (Definition 2.2 (S3)).

(c2) *As a theorem constructor*: the witness is supplied by a named claim of a submodule of this module, which constructs one channel for each member of a declared family of descriptors and proves the identity of (c1) for it. The identity is then *not* decided on numbers: the link carries an *obligation record* of four components, the triple

$$\big(\mathsf{kind},\ \text{the id of the soundness claim},\ \mathcal{A}(\mathsf{kind})\big) \tag{2.2}$$

and, as the fourth, the member of the declared family the link's descriptors are. The third component is the assumptions the obligation

carries, which Definition 2.2 (A1) defines as the soundness claim's assumption closure together with the physical-source assumption the submodule declares for the source model the claim is stated over, an assumption the soundness claim itself does *not* assume. The obligation record travels with the link and, where the link carries a licence record, in it. *Its first three components are decided*, against the soundness registry of the artefact that carries the link (Definition 2.2 (A1), (A2)), by a comparison of declared strings. The fourth is not: which member of the family the descriptors are is asserted by the module that emits the link and re-proved by no consumer, and no verdict on a number is carried, there being none to decide.

A channel supplied as a theorem constructor is how a link whose source or target is not a matrix is expressed at all, and it is the only way an obligation enters this module: no claim here discharges one.

(d) The declared descriptor family. $\mathcal{D}_\Xi$, the *declared descriptor family*, is the family of source descriptors the witness's identity is asserted for. It contains the declared source descriptor; for a link whose channel is supplied as data it is that descriptor alone; for a constructor link it is the family the named claim proves the identity for. The link is *sound* when $\Gamma_\Xi(\sigma_x^{\mathrm{in}}) = \sigma_x^{\mathrm{out}}$ holds for every $x \in \mathcal{X}$ at every member of $\mathcal{D}_\Xi$; for a data link this is what (S3) decides, and for a constructor link it is what the named claim concludes and the obligation record names. A link whose family does not contain its own declared descriptor asserts nothing about the descriptor it carries, and is refused. For a constructor link that membership is not a decidable condition: the family is the extension of a hypothesis of a claim, the emitting module asserts that its descriptors satisfy it, and what a consumer decides instead is the triple (2.2) against the registry and the provenance of the artefact (Definition 2.2 (A1) to (A3)). The refusal of a family that excludes its own descriptor therefore applies where membership is decidable, to a data link, whose family is by definition its own descriptor, so that it refuses a data link whose family slot excludes that descriptor; elsewhere membership is an obligation of the emitting module, and what the registry bounds is which claims a link may name for it.

(e) The retained-register sector map. $\mathfrak{s}_\Xi$, the *retained-register sector map*, is optional: a link may emit a matched pair of sector structures over one ordered index set, one on Alice's retained system A and one on the Alice-side descriptor *the certificate is written against*, which is the *source* descriptor of the segment's first link, the virtual pair (Definition 2.2 (S1)), in the sense the block form of Definition 3.3 is written against. It is not the descriptor the link produces: a link's target may be a system that carries no sector structure at all, and for a constructor link whose target is infinite dimensional it never carries one. The sector map is a presentation datum: it says how the certificate, the marginal and Alice's measurement are presented, and it is no

statement about any joint state (Definition 1.1, "What the sector structure is not"). Where the slot is empty the one-sector structure is meant.

(f) The transported constraint entries and the licence. $\mathfrak{L}_\Xi$ is the list of *transported constraint entries*, entries of a consumer's constraint list carried across the link, each carrying, *per entry*, the certificate case that discharges it — the statement, for that entry, that what the entry asserts of the link's target is implied by what holds of its source — and $\mathsf{licence}$ is the link's licence record, the slot every link of a chain has for its transported entries (Definition "Typed link" (reductions/def:link)), present exactly when $\mathfrak{L}_\Xi \neq \varnothing$. *A non-empty list is refused*: a consumer's constraint operators live on the measured system, and no claim of this repository transports an Alice-side entry. An admitted link therefore carries an empty list and no licence record of its own.

(g) The status. $\mathsf{status} \in \{\mathsf{exact}, \mathsf{certified_outer}, \mathsf{exploratory_numeric}\}$, the link statuses of Definition "Typed link" (reductions/def:link), with exact meaning that the identity of (c1) holds with no error term — for a constructor link, that the named claim concludes it with none.

Alice-side descriptor discipline. Both descriptors of a source link are Alice-side: each names a transmitted system and an ensemble on it, and no condition on a source link mentions the receiver's systems, its outcome alphabets or its measurements. A source link changes what Alice emits and nothing else.

The degenerate link. A link that supplies its channel as data, whose declared descriptor family is that one descriptor, whose sector-map slot is empty and whose entry list is empty gains nothing from its slots (d) and (e): they carry exactly what the identity of (c1) asserts and the one-sector structure.

Definition 2.2 (Source segment and its interface conditions)**.** The *source segment* of a reduction chain is an ordered list $\mathfrak{R}_{\mathrm{src}} = (\Xi_1, \ldots, \Xi_{L_\mathrm{s}})$ of source links (Definition 2.1), $L_\mathrm{s} \geq 0$, whose links act on the transmitted system and replace Alice's signal ensemble; it is presented together with a source specification $\mathfrak{S}_{\mathrm{src}}$ (Definition 1.1). The segment satisfies the *interface conditions* when the following hold.

(S1) *(descriptor match)* the target descriptor of Ξ_i equals the source descriptor of Ξ_{i+1} for $1 \leq i < L_\mathrm{s}$; the target descriptor of Ξ_{L_s} is the specified pair, the transmitted system A' named by $\mathfrak{S}_{\mathrm{src}}$ with the specified ensemble $\{p_x, \rho_x\}$ on it; and the source descriptor of Ξ_1 is the virtual pair, the *virtual transmitted system* A'' with the *virtual ensemble* $\{p_x, \rho_x^{\mathrm{v}}\}_{x \in \mathcal{X}}$ on it, on which the source-replacement certificate is written (Definition 3.3). Every descriptor of this segment is an Alice-side descriptor: no condition of the segment mentions the receiver's systems, its outcome alphabets or its measurements;

(S2) *(index preservation)* each link carries the signal index x through unchanged: it maps a family indexed by $\mathcal{X}$ to a family indexed by $\mathcal{X}$, with the weights p_x untouched;

(S3) *(source identity, where checkable)* writing $\Xi := \Xi_{L_s} \circ \cdots \circ \Xi_1$ for the composite when every link supplies its channel as data,

$$\Xi(\rho_x^{\mathrm{v}}) = \rho_x \qquad \text{for every } x \in \mathcal{X}, \tag{2.3}$$

an *exact* matrix identity, decided entry by entry. The composite Ξ is *formed* in exact arithmetic as well, so that the identity decided is the identity written; that the operands are binary64 is not what makes the decision exact (the number model of Definition 1.1). The channel runs from the *virtual* states to the *specified* ones, and the object a consumer receives is the virtual protocol, the protocol with the virtual ensemble in the place of the specified one. *Where a link supplies its channel as a theorem constructor* (Definition 2.1 (c2)) the identity of *that* link is not decided on numbers: it is the conclusion of the claim the link's obligation record names, at the member of that claim's declared descriptor family which the link's descriptors are, and the identity of the composite is the composition of the per-link identities — each one either decided, where its link supplies data, or concluded by its named claim, where its link supplies a constructor. In a segment that mixes the two kinds of witness, each data link's identity is therefore decided link by link. So (2.3) is what the composite satisfies in every case, and what changes with the kind of witness is *by what* it is established and therefore what a consumer may re-decide: nothing of a constructor link's identity, which is why the obligation travels;

(S4) *(slots)* each link carries its constraint entries and its licence slot, both possibly empty, and its status tag; a link whose channel is a theorem constructor carries in addition its obligation record, the claim and the assumptions it names, which are the strings (A2) below decides.

Which kinds are admitted, and what a consumer decides. No source link kind is admitted unless the submodule that declares it supplies its soundness claim: a link of kind k may appear in a segment only where a submodule of this module declares k and supplies, as a claim of this repository, the statement that the link's witness satisfies the identity of Definition 2.1 (c1) at every member of its declared descriptor family — decided there, for a data link, or proved there, for a constructor link. A link of a kind for which no such claim exists is *refused*, and so is a link with a non-empty list of transported constraint entries, whatever its kind: no claim of this repository transports an Alice-side entry.

That much is a condition on the *kind*, and a consumer cannot decide it of a claim it does not read. What a consumer decides is the following, and it is decided on data.

(A1) *(the soundness registry)* this module emits, with every artefact, a finite list — the *soundness registry* — with one entry per admitted source-link kind, *closed* in that a kind without an entry is not admitted (Definition 2.1 (a)), each entry the triple

$$\big(\mathsf{kind},\ \text{the id of its soundness claim},\ \mathcal{A}(\mathsf{kind})\big), \tag{2.4}$$

written out in the module's interface and emitted verbatim. *The third component $\mathcal{A}(\mathsf{kind})$ is the assumptions the obligation carries*, and it has two sources, both of which a registry entry lists:

(a) the *assumption closure of the soundness claim*: the assumptions in the transitive closure of that claim under its *uses* and *assumes* edges, the claims and assumptions each claim lists as its dependencies; it may be empty, and is empty if and only if no claim of that closure, the soundness claim included, lists an assumption among its dependencies;

(b) the *physical-source assumption the submodule declares for the source model the claim is stated over*, where the claim is stated over one. That assumption is not a premise of the soundness claim and must not be made one: what it attests is that the model the claim is about describes the physical device, which is what an obligation on a physical source owes and a mathematical identity does not.

The registry of an artefact is the registry of the version of this module that emitted it, not a datum a request supplies.

(A2) *(the triple is decided)* a presented chain's link is *refused* unless the triple (its kind, the claim id of its obligation record, the assumption short ids of that record) appears in the registry of the artefact that carries it — a comparison of finitely many declared strings, taken entry by entry — and unless every assumption of that entry is in the artefact's assumption ledger, the list of assumptions the artefact rests on. *Both components of $\mathcal{A}(\mathsf{kind})$ are compared*, so a link that names the soundness claim's closure and omits the model's attestation is refused exactly as one that names neither. A link supplying its channel as data carries no obligation record and is decided by (S3) instead.

(A3) *(provenance)* a non-empty source segment is accepted only inside an artefact emitted by the module that admits its links. A segment assembled from links a consumer received in any other way is refused,

whatever its triples: the registry says which kinds this repository has proved, not that a particular link was built by the module that proved them.

The admission rule on the kind is applied where a link is admitted, which is this module and the submodule whose kind it is; it is not among the conditions (S1) to (S4), which are the conditions a consumer decides on a presented chain, and (A1) to (A3) are conditions on the artefact and its registry rather than on the segment's descriptors.

The empty segment. When $L_{\mathrm{s}} = 0$ the segment is empty, Ξ is the identity, $A'' = A'$ and $\rho_x^{\mathrm{v}} = \rho_x$, so (2.3) holds by inspection. That is the case of every specification whose ensemble is declared as matrices and that carries no link. A specification whose ensemble is declared by a source model is presented with a non-empty segment (Definition 1.1, case (b)), which is the other case and the one the conditions above are written for.

3 The source-replacement certificate

Fact 3.1 (Source replacement for prepare-and-measure protocols)**.** *([1, Lemma 8.3, Eqs. (83)–(84)], with the protocol framing of its Def. 8.1 and Def. 8.2.)* Consider a protocol in which Alice first prepares the global state $\rho_{X_1^n (A')_1^n} = \bigotimes_{j=1}^{n} \rho^{(j)}_{X_j A'_j}$ with $\rho^{(j)}_{X_j A'_j} = \sum_{k=1}^{d_A} p_k^{(j)} \, |k\rangle\langle k|_{X_j} \otimes (\rho_k^{(j)})_{A'_j}$, where $\{|k\rangle\}_{X_j}$ is the classical basis of X_j; she keeps X_1^n and sends the $(A')_1^n$ systems to the adversary. Then the state $\rho_{X_1^n (A')_1^n}$ can equivalently be obtained by *(1)* preparing the global state $\rho_{\bar{A}_1^n \hat{A}_1^n (A')_1^n} = \bigotimes_j \rho^{(j)}_{\bar{A}_j \hat{A}_j A'_j}$ with $\rho^{(j)}_{\bar{A}_j \hat{A}_j A'_j} = \sum_{k,i=1}^{d_A} \sqrt{p_i^{(j)} p_k^{(j)}} \; |i\rangle\langle k|_{\bar{A}} \otimes \left|\psi_i^{(j)}\right\rangle\!\left\langle\psi_k^{(j)}\right|_{\hat{A}_j A'_j}$, where $\left|\psi_k^{(j)}\right\rangle_{\hat{A}_j A'_j}$ is a purification of $(\rho_k^{(j)})_{A'_j}$, $\{|k\rangle\}$ is an orthonormal basis of $\bar{A}$ and $\hat{A}_j$ is the *shield system*; *(2)* sending $(A')_1^n$ to the adversary; and *(3)* for each j, measuring $\bar{A}_j \hat{A}_j$ with the POVM $\{|k\rangle\langle k|_{\bar{A}_j} \otimes \mathbb{1}_{\hat{A}_j}\}_{k \in \{1,\ldots,d_A\}}$ and storing the result in X_j.

Symbol map and hypothesis check. The source's X_j is this module's outcome register with alphabet $\mathcal{X}$ and $k \leftrightarrow x$; its A'_j is the transmitted system carrying the ensemble the lemma is applied to, and its $\{p_k^{(j)}, \rho_k^{(j)}\}$ is that ensemble. That ensemble is the virtual pair $(A'', \{p_x, \rho_x^{\mathrm{v}}\})$ on which the source-replacement certificate is stated, which is the specified pair $(A', \{p_x, \rho_x\})$ of Definition 1.1 exactly when the source segment is empty. This module describes one round; that every round is described by the same data is the consumer's round-homogeneity assumption and is not asserted here, and it is under that assumption that the superscript j is dropped in what follows. The source's $\bar{A}_j \hat{A}_j$ is this module's retained system A, and its measurement POVM is $\{M_x\}$ with $M_x = |x\rangle\langle x|_{\bar{A}} \otimes \mathbb{1}_{\hat{A}}$. The d_A of the cited lemma and

the d_A of this module are *not* the same number, and the identification does not claim they are: the published index runs $k \in \{1, \ldots, d_A\}$ over an orthonormal basis of $\bar{A}$, so there $d_A = \dim \mathcal{H}(\bar{A}) = |\mathcal{X}|$, whereas throughout this module $d_A = \dim \mathcal{H}(A) = \dim \mathcal{H}(\bar{A}) \cdot \dim \mathcal{H}(\hat{A})$; the two agree exactly when the shield system is trivial. The lemma's one hypothesis is that Alice prepares the classical-quantum source state of its Eq. (83) — this is Assumption 5.1 for the physical device, and the declared source data for the specified object. Its only timing element is that the global state is prepared first; it places no condition on the sequential structure of later operations, being an identity between the state of its Eq. (83) and the measured state of its Eq. (84). The cited work applies it within the protocol framing of its Def. 8.1 and Def. 8.2, and this module applies it to protocols with the sequential structure of Assumption 5.2, which supplies that structure: the assumption is the hypothesis under which this module applies the fact, and not a hypothesis of the cited lemma. The published statement quantifies over an arbitrary purification $|\psi_k\rangle$, so the shield system is trivial exactly when every state of that ensemble is pure — the case, for a pure *specified* ensemble, of [2, Eqs. (37)–(39)], which writes the source-replaced vector as $|\psi\rangle_{\bar{A}A'} = \sum_i \sqrt{p_i}\, |i\rangle_{\bar{A}}\, |\varphi_i\rangle_{A'}$ and records that the resulting marginal $\rho_{\bar{A}} = \mathrm{tr}_{A'}(|\psi\rangle\langle\psi|)$ is fixed independently of the adversary's attack.

Definition 3.3 (Source-replacement certificate)**.** Let $\{p_x, \rho_x^{\mathrm{v}}\}_{x \in \mathcal{X}}$ be the *virtual* signal ensemble on the virtual transmitted system A'' — the pair the source segment starts from, Definition 2.2 (S1), (S3) — with weighted members $\sigma_x^{\mathrm{v}} := p_x \rho_x^{\mathrm{v}} \in \mathrm{Pos}(A'')$, $p_x = \mathrm{tr}\, \sigma_x^{\mathrm{v}} \geq 0$ and $\sum_x p_x = 1$. These conditions are decided *exactly*, as every declared condition is, the weight sum in the declared order and rounding nowhere. A *source-replacement certificate* for the declared data $(A, \mathbb{T}, \tau_A, \{M_x\})$ of a source specification (Definition 1.1) is an operator $\Psi \in \mathcal{L}(AA'')$ such that

(SR1) $\mathrm{tr}_{A''}\, \Psi = \tau_A$;

(SR2) $\mathrm{tr}_A\big[(M_x \otimes \mathbb{1}_{A''})\, \Psi\big] = p_x\, \rho_x^{\mathrm{v}}$ for every $x \in \mathcal{X}$; and

(SR3) *(retention, an attestation)* A is retained in Alice's laboratory and never transmitted; no operation of the protocol and no operation of the adversary acts on A, except Alice's declared measurement $\{M_x\}$.

It is given in one of the three forms below. In the pure form and in the block form Ψ is declared, together with the declared computational basis of A. In an ensemble presentation neither Ψ nor a basis of A is declared: Ψ is defined from a declared ensemble by (3.2).

Pure form. In the *pure form* the sector structure of A is the one-sector structure, and Ψ is a *rank-one projector* — Ψ Hermitian, $\Psi^2 = \Psi$ and $\mathrm{tr}\, \Psi = 1$, each decided exactly and the trace summed without rounding, so that $\Psi = |\psi\rangle\langle\psi|$ for a unit vector $|\psi\rangle \in \mathcal{H}(AA'')$.

Block form. Let $\mathcal{H}(A) = \bigoplus_{t\in\mathbb{T}} \mathcal{H}(A_t)$ be the sector structure of A declared in Definition 1.1, along which $\tau_A = \bigoplus_t \tau_A^{(t)}$ and $M_x = \bigoplus_t M_x^{(t)}$ are presented as block lists. In the *block form* the virtual transmitted system carries a declared sector structure *matched* to that of A: the same ordered index set $\mathbb{T}$ and a decomposition $\mathcal{H}(A'') = \bigoplus_{t\in\mathbb{T}} \mathcal{H}(S_t)$ into subspaces each spanned by vectors of the declared basis of A'', with projectors Π_t^{v}. The certificate is a block list $(\Psi_t)_{t\in\mathbb{T}}$ with $\Psi_t \in \mathcal{L}(A_tS_t)$, each block Hermitian and positive semidefinite, decided exactly, and $\sum_t \operatorname{tr}\Psi_t = 1$, the traces summed in the declared order without rounding. It denotes the operator $\Psi = \bigoplus_t \Psi_t$ on the *matched subspace* $\bigoplus_t \mathcal{H}(A_t)\otimes\mathcal{H}(S_t) \subseteq \mathcal{H}(AA'')$, extended by zero to its orthogonal complement. No purity condition is imposed, on Ψ or on any block. The weighted virtual members are required to be *block diagonal* along $(S_t)_t$: $\Pi_t^{\mathrm{v}}\sigma_x^{\mathrm{v}}\Pi_{t'}^{\mathrm{v}} = 0$ for all $t \neq t'$ and every x, that is, every entry of σ_x^{v} whose row lies in one sector and whose column lies in another is zero, decided entry by entry on the declared entries of σ_x^{v} (a virtual ensemble declared as block lists along $(S_t)_t$ meets it by its presentation). Write $\sigma_x^{\mathrm{v},(t)} \coloneqq \Pi_t^{\mathrm{v}}\sigma_x^{\mathrm{v}}\Pi_t^{\mathrm{v}}$, read as an operator on S_t, for its diagonal blocks. Against a block list the two identities read block by block,

$$\operatorname{tr}_{S_t}\Psi_t = \tau_A^{(t)}, \qquad \operatorname{tr}_{A_t}\big[(M_x^{(t)}\otimes\mathbb{1}_{S_t})\,\Psi_t\big] = \sigma_x^{\mathrm{v},(t)} \qquad \text{for every } t\in\mathbb{T},\ x\in\mathcal{X}, \tag{3.1}$$

and (3.1) together with the block diagonality of every σ_x^{v} is equivalent to (SR1) and (SR2) of the operator Ψ. Indeed Ψ vanishes off the matched subspace and M_x is block diagonal, so the partial traces of Ψ are the direct sums of those of its blocks: $\operatorname{tr}_{A''}\Psi = \bigoplus_t \operatorname{tr}_{S_t}\Psi_t$, which is $\tau_A = \bigoplus_t \tau_A^{(t)}$ exactly when the first identity of (3.1) holds; and $\operatorname{tr}_A[(M_x\otimes\mathbb{1}_{A''})\Psi] = \bigoplus_t \operatorname{tr}_{A_t}[(M_x^{(t)}\otimes\mathbb{1}_{S_t})\Psi_t]$, which is block diagonal along $(S_t)_t$ and so equals σ_x^{v} exactly when σ_x^{v} is block diagonal and its diagonal blocks satisfy the second identity. Without the block diagonality, (3.1) is (SR2) against the pinched members $\bigoplus_t \sigma_x^{\mathrm{v},(t)}$ and not against σ_x^{v}.

Ensemble presentation. A block-form certificate need not be declared at all. In an *ensemble presentation* the declared data are the matched sector structure $(S_t)_{t\in\mathbb{T}}$ on A'' and the weighted virtual ensemble as block lists, $\sigma_x^{\mathrm{v}} = \bigoplus_t \sigma_x^{\mathrm{v},(t)}$ with $\sigma_x^{\mathrm{v},(t)} \in \mathcal{L}(S_t)$, and *nothing else*: no Ψ, no $\{M_x\}$, no τ_A and no basis of A. The conditions are

(E1) every block $\sigma_x^{\mathrm{v},(t)}$ is Hermitian and positive semidefinite, decided exactly;

(E2) $p_x = \sum_t \operatorname{tr}\sigma_x^{\mathrm{v},(t)}$ for every x and $\sum_x p_x = 1$, the sums formed in the declared orders and rounding nowhere; and

(E3) for every t the sum $\sum_{x\in\mathcal{X}} \sigma_x^{\mathrm{v},(t)}$, accumulated entry by entry in the declared order of $\mathcal{X}$, rounds nowhere, so that its entries are binary64

numbers.

The system A, its sector structure, the marginal τ_A, the POVM $\{M_x\}$ and the blocks Ψ_t are then *defined* by (3.2): A_t is a copy of S_t, $\tau_A^{(t)}$ is the transpose of the sum in (E3), and Ψ_t and $M_x^{(t)}$ are the canonical purification and its square-root measurement. By Lemma 3.4 they are a block-form certificate with (SR1) and (SR2) holding as identities, with no error term, against the declared ensemble itself, and with every non-zero block a multiple of a rank-one projector. (E1) to (E3) are all that is decided. (SR1), (SR2), positivity of the blocks, the POVM condition and the purity of the blocks are *theorems* of the presentation and are decided by nothing, there being no second declared datum they could be compared with; (SR3) is the attestation it is in every form. Of the defined objects only τ_A is ever formed as numbers, and by (E3) it is admissible declared data. Ψ_t and $M_x^{(t)}$ carry square roots and quotients of the declared numbers, are formed by no one, and are never rounded.

What a consumer reads of a presentation. What a consumer can read of Alice's measurement without Ψ_t and $M_x^{(t)}$ is the family $N_x^{(t)} = (\sigma_x^{\mathrm{v},(t)})^{\mathsf{T}}$ of Lemma 3.4 (v). *A consumer whose declaration carries Alice's measurement as the POVM $\{M_x\}$ cannot receive an ensemble presentation and refuses it*; the form of declaration that carries $\{N_x\}$ in its place is that consumer's to define, and the consuming modules define it: the protocol compiler and the key-rate engine receive a presentation through the ensemble form of their declarations, which carries $\{N_x\}$ in the place of $\{M_x\}$ (Definition "Protocol declaration" (protocol-compiler/def:declaration), Definition "The protocol declaration as the MEAT core reads it" (meat-core/def:declaration)), admitted beside a presentation by Definition "Key-length eligibility" (protocol-compiler/def:eligible). The mode, the alphabet, the source segment and the attestations of a specification are what they are in the other forms.

What is decided. (SR1) and (SR2) are *exact matrix identities* on declared data, decided entry by entry. The conditions a consumer decides — in pure form the three on Ψ, (SR1) and (SR2); in block form Hermiticity and positive semidefiniteness of every block, the trace, the block diagonality of every σ_x^{v} along $(S_t)_t$, and (3.1), which together are (SR1) and (SR2); in an ensemble presentation (E1) to (E3) and no others — are stated in these terms and in no others: none of them is stated on a vector, which is not a declared datum. (SR3) carries no number and is decided by nothing; it is the attestation of Assumption 5.3. A rank-one projector of unit trace is positive semidefinite of unit trace, so a pure-form certificate satisfies every condition of the block form at the one-sector structure, and one condition more.

One reading at every L_{s}. The pair $(A'', \{p_x, \rho_x^{\mathrm{v}}\})$ is the only pair a certificate is ever written against, at every length of the source segment, because the object a consumer receives is the virtual protocol (Definition 2.2 (S3)): the specification's τ_A and $\{M_x\}$ describe the preparation of $\{p_x, \rho_x^{\mathrm{v}}\}$, and it

is the source identity (2.3), not the certificate, that carries that description back to the specified ensemble $\{p_x, \rho_x\}$ on A'. At $L_{\mathrm{s}} = 0$ the two pairs are literally equal, $A'' = A'$ and $\rho_x^{\mathrm{v}} = \rho_x$. So (SR1)–(SR2) have exactly one reading whatever L_{s} is.

Lemma 3.4 (Certificate defined by an ensemble presentation). Let the virtual transmitted system carry a sector structure $\mathcal{H}(A'') \supseteq \bigoplus_{t\in\mathbb{T}} \mathcal{H}(S_t)$, each S_t spanned by vectors $(|i\rangle_{S_t})_i$ of the declared basis, and let $(\sigma_x^{\mathrm{v},(t)})_{t\in\mathbb{T}}$, $x \in \mathcal{X}$, be block lists with $\sigma_x^{\mathrm{v},(t)} \in \mathrm{Pos}(S_t)$ and $\sum_x \sum_t \operatorname{tr} \sigma_x^{\mathrm{v},(t)} = 1$. For every t let A_t be a copy of S_t with basis $(|i\rangle_{A_t})_i$ in the same order, let transposition $X \mapsto X^{\mathsf{T}}$ be taken in these bases, and put

$$\begin{aligned} \tau_S^{(t)} &:= \sum_{x\in\mathcal{X}} \sigma_x^{\mathrm{v},(t)}, \qquad \tau_A^{(t)} := \big(\tau_S^{(t)}\big)^{\mathsf{T}}, \\ |\psi_t\rangle &:= \sum_i |i\rangle_{A_t} \otimes \big(\tau_S^{(t)}\big)^{1/2} |i\rangle_{S_t}, \qquad \Psi_t := |\psi_t\rangle\langle\psi_t|, \\ M_x^{(t)} &:= \big(\tau_A^{(t)}\big)^{-1/2} \big(\sigma_x^{\mathrm{v},(t)}\big)^{\mathsf{T}} \big(\tau_A^{(t)}\big)^{-1/2} + \delta_{x,x_0} \big(\mathbb{1}_{A_t} - P_t\big), \end{aligned} \tag{3.2}$$

where P_t is the projector onto the support of $\tau_A^{(t)}$, the inverse square root is taken on that support and is zero off it, and x_0 is the first letter of $\mathcal{X}$ in its declared order. Then, with $A := \bigoplus_t A_t$:

(i) *(a certificate in block form)* every Ψ_t is positive semidefinite, $\sum_t \operatorname{tr} \Psi_t = 1$, $\{M_x\}_x$ with $M_x = \bigoplus_t M_x^{(t)}$ is a POVM presented as a block list, $\tau_A = \bigoplus_t \tau_A^{(t)}$ is a state presented as a block list, and both identities of (3.1) hold: $(\Psi_t)_t$ satisfies every condition Definition 3.3 places on a block-form certificate for $(A, \mathbb{T}, \tau_A, \{M_x\})$ against the virtual ensemble with weighted members $\sigma_x^{\mathrm{v}} = \bigoplus_t \sigma_x^{\mathrm{v},(t)}$, (SR3) apart, which is an attestation;

(ii) *(pure blocks)* every non-zero Ψ_t is $g_t := \operatorname{tr} \tau_S^{(t)}$ times a rank-one projector, so the converse of Theorem 4.1 (ii) holds of this certificate;

(iii) *(what is formed as numbers)* the entry (i, j) of $\tau_A^{(t)}$ is the entry (j, i) of the sum $\tau_S^{(t)}$ of the declared blocks. No other object of (3.2) enters (i) or (ii) through its entries: Ψ_t and $M_x^{(t)}$ contain square roots and quotients of the declared numbers and are, in general, not declarable;

(iv) *(the completion is immaterial)* the letter x_0 enters the POVM condition and nothing else: (SR1), (SR2), (ii) and (v) hold with the term $\mathbb{1}_{A_t} - P_t$ assigned to any letter, or split among the letters by any positive semidefinite operators on the kernel of $\tau_A^{(t)}$ that sum to $\mathbb{1}_{A_t} - P_t$;

(v) *(Alice's side without a square root)* $\big(\tau_A^{(t)}\big)^{1/2} M_x^{(t)} \big(\tau_A^{(t)}\big)^{1/2} = \big(\sigma_x^{\mathrm{v},(t)}\big)^{\mathsf{T}}$ for every x and t: the operators $N_x^{(t)} := (\sigma_x^{\mathrm{v},(t)})^{\mathsf{T}}$ are entrywise the declared numbers, and $\sum_x N_x^{(t)} = \tau_A^{(t)}$.

4 Sufficiency of the certificate and its scope

Theorem 4.1 (Checkability and sufficiency of the source-replacement certificate). Let a source specification (Definition 1.1) carry a source-replacement certificate Ψ for $(A, \mathbb{T}, \tau_A, \{M_x\})$ against the virtual ensemble $\{p_x, \rho_x^{\mathrm{v}}\}$, in any form of Definition 3.3. In (ii) and (iii) and their proofs an ensemble presentation is read as the block-form certificate it defines (Lemma 3.4 (i)), so that "either form" below covers it.

(i) *(checkability)* the conditions on Ψ of its declared form — in pure form Hermiticity, $\Psi^2 = \Psi$ and $\operatorname{tr}\Psi = 1$; in block form Hermiticity and positive semidefiniteness of every block and $\sum_t \operatorname{tr}\Psi_t = 1$ — together with (SR1) and (SR2), are finitely many exact identities and exact decisions on the declared data, decided by finitely many arithmetic operations. In block form (SR1) and (SR2) are decided as the block identities (3.1) together with the block diagonality of every weighted virtual member σ_x^{v} along $(S_t)_t$, decided entry by entry, which by Definition 3.3 is their conjunction; (3.1) alone decides (SR2) only against the pinched members $\bigoplus_t \Pi_t^{\mathrm{v}} \sigma_x^{\mathrm{v}} \Pi_t^{\mathrm{v}}$. For a certificate given by an ensemble presentation the decided conditions are (E1) to (E3) of Definition 3.3, equally finite and exact, and the conditions of the block form hold of the defined certificate by Lemma 3.4 (i) without being decided. (SR3) is an attestation and is decided by no function of the declared data. (SR1) moreover implies $\tau_A \in S_=(A)$, $\operatorname{tr}_{A''}\Psi$ being positive semidefinite with trace $\operatorname{tr}\Psi = 1$. In an ensemble presentation τ_A is not declared: it is formed as the transposed sum of (E3), and that it is a state is Lemma 3.4 (i).

(ii) *(consistency)* more generally than in a source specification, let $\{M_x\}$ be any family of Hermitian operators on A, presented as block lists against the sector structure, of which $\sum_x M_x = \mathbb{1}_A$ is not assumed; let Ψ meet the conditions on Ψ of its form listed in (i) and (SR1); and let $p_x\rho_x^{\mathrm{v}}$ stand for $\operatorname{tr}_A[(M_x \otimes \mathbb{1}_{A''})\Psi]$, so that (SR2) holds. Writing $U = \operatorname{supp}\tau_A$ with projector Π_U, the certificate's signal-side marginal and the virtual average signal state agree,

$$\sum_x p_x \rho_x^{\mathrm{v}} = \operatorname{tr}_A \Psi, \tag{4.1}$$

if $\Pi_U\big(\sum_x M_x\big)\Pi_U = \Pi_U$, in either form; in particular they agree for the certificate of a source specification, whose $\{M_x\}$ is a POVM by Definition 1.1. The converse, *only if*, holds in pure form, and in block form whenever every non-zero block Ψ_t is a positive multiple of a rank-one projector; it fails for a general positive block.

(iii) *(sufficiency at the module boundary)* Assume (SR1), (SR2), the attestation (SR3) of Assumption 5.3, and Assumption 5.2; in block form

(SR2) is the identity of operators on A'' against the weighted virtual members $p_x \rho_x^{\mathrm{v}}$ themselves, and not against their diagonal blocks alone: for a declared block form (i) decides it with their block diagonality, and for an ensemble presentation it is proved by Lemma 3.4 (i). Let $\mathcal{N}_{\mathsf{A}}$ be Alice's measurement map, from her retained system A to the classical outcome register X,

$$\mathcal{N}_{\mathsf{A}}(\xi) := \sum_{x \in \mathcal{X}} \mathrm{tr}[M_x\, \xi]\ |x\rangle\langle x|_X\,, \qquad \xi \in \mathcal{L}(A), \tag{4.2}$$

and let $\mathcal{E}$ be *any* channel from $A''E$ to E', where E is the adversary's initial register and E' collects her output registers including whatever is delivered to the receiver. Thus $\mathcal{E}$ models the adversary's attack and the remainder of the round *on the transmitted system and the adversary's registers*, and it acts on neither A nor X: it is not permitted to depend on Alice's outcome. Write $\rho_{XA''} := \sum_{x \in \mathcal{X}} p_x\ |x\rangle\langle x|_X \otimes \rho_x^{\mathrm{v}}$ for the virtual prepare-and-measure source state. Then:

(a) *(the interchange)*

$$(\mathrm{id}_X \otimes \mathcal{E}) \circ (\mathcal{N}_{\mathsf{A}} \otimes \mathrm{id}_{A''E}) \;=\; \mathcal{N}_{\mathsf{A}} \otimes \mathcal{E} \;=\; (\mathcal{N}_{\mathsf{A}} \otimes \mathrm{id}_{E'}) \circ (\mathrm{id}_A \otimes \mathcal{E}) \tag{4.3}$$

as linear maps on $\mathcal{L}(AA''E)$, hence in particular on $\Psi \otimes \sigma$ for every state σ on E: Alice's measurement may be performed before the attack or after it, indifferently. This uses no property of the certificate, only that the two maps act on complementary registers;

(b) *(the source identification)* for every state σ on E,

$$(\mathcal{N}_{\mathsf{A}} \otimes \mathcal{E})\big(\Psi \otimes \sigma\big) \;=\; (\mathrm{id}_X \otimes \mathcal{E})\big(\rho_{XA''} \otimes \sigma\big), \tag{4.4}$$

and both sides are states on XE'. Together with (a): on every register other than A, the declared entanglement-based description and the virtual prepare-and-measure description produce the same state, against every attack of this form. The states ρ_x^{v} in $\rho_{XA''}$ are the declared virtual states, in block form as in every form, and not their pinching along $(S_t)_t$.

Corollary 4.3 (Certificates beyond the canonical form)**.** Throughout, the certificate is in the *pure* form of Definition 3.3, against the one-sector structure on A that Definition 1.1 makes the default.

(i) Fact 3.1's canonical data — $A = \bar{A}\hat{A}$ with $\dim \mathcal{H}(\bar{A}) = |\mathcal{X}|$, $M_x = |x\rangle\langle x|_{\bar{A}} \otimes \mathbb{1}_{\hat{A}}$, and $\Psi = |\psi\rangle\langle\psi|$ with $|\psi\rangle = \sum_x \sqrt{p_x}\, |x\rangle_{\bar{A}}\, |\psi_x\rangle_{\hat{A}A''}$ and $|\psi_x\rangle$ a purification of ρ_x^{v} — satisfy (SR1) and (SR2).

(ii) They are not the only such data: Theorem 4.1 (iii) applies to every triple $(\Psi, \tau_A, \{M_x\})$ satisfying (SR1)–(SR3), and there are source spec-

ifications with $d_A < |\mathcal{X}|$ carrying a certificate. In particular the four-state BB84 ensemble on a qubit, $\mathcal{X} = \{(b,k) : b,k \in \{0,1\}\}$ with $p_{(b,k)} = \frac{1}{4}$ and $\rho^{\mathrm{v}}_{(b,k)} = |k_b\rangle\langle k_b|$ for the computational basis ($b = 0$) and the Hadamard basis ($b = 1$), admits the certificate $A = A'' = \mathbb{C}^2$, $M_{(b,k)} = \frac{1}{2}|k_b\rangle\langle k_b|$, $\tau_A = \frac{1}{2}\mathbb{1}_A$, $\Psi = |\Phi^+\rangle\langle\Phi^+|$ with $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$: $d_A = 2$ where the canonical form has $d_A = 4$.

Lemma 4.5 (Matched pinching of a block certificate). Let a source specification carry a certificate in block form (Definition 3.3), with sector projectors Π^A_t on A and Π^{v}_t on A'', $t \in \mathbb{T}$, which sum to the identity on A and on A'' respectively (Definitions 1.1 and 3.3), and write

$$\mathrm{Pin}_A(\xi) \coloneqq \sum_{t\in\mathbb{T}} \Pi^A_t\, \xi\, \Pi^A_t, \qquad \mathrm{Pin}_{\mathrm{v}}(\zeta) \coloneqq \sum_{t\in\mathbb{T}} \Pi^{\mathrm{v}}_t\, \zeta\, \Pi^{\mathrm{v}}_t$$

for the two pinching channels.

(i) *(matched structure)* for every $t \in \mathbb{T}$,

$$(\Pi^A_t \otimes \mathbb{1}_{A''})\,\Psi = \Psi\,(\Pi^A_t \otimes \mathbb{1}_{A''}) = (\mathbb{1}_A \otimes \Pi^{\mathrm{v}}_t)\,\Psi = \Psi\,(\mathbb{1}_A \otimes \Pi^{\mathrm{v}}_t) = \Psi_t, \tag{4.5}$$

the block Ψ_t read as an operator on $\mathcal{H}(AA'')$ extended by zero. Hence $(\mathrm{Pin}_A \otimes \mathrm{id}_{A''})(\Psi) = \Psi = (\mathrm{id}_A \otimes \mathrm{Pin}_{\mathrm{v}})(\Psi)$.

(ii) *(reached states)* assume (SR3), and let $\mathcal{E}$ be any channel from $A''E$ to E' and σ any state on E, as in Theorem 4.1 (iii). The state $\omega \coloneqq (\mathrm{id}_A \otimes \mathcal{E})(\Psi \otimes \sigma)$ on AE' satisfies

$$\mathrm{tr}_{E'}\,\omega = \tau_A, \qquad (\mathrm{Pin}_A \otimes \mathrm{id}_{E'})(\omega) = \omega, \qquad \omega = (\mathrm{id}_A \otimes \mathcal{E}')(\Psi \otimes \sigma) \;\text{ with }\; \mathcal{E}' \coloneqq \mathcal{E} \circ (\mathrm{Pin}_{\mathrm{v}} \otimes \mathrm{id}_E). \tag{4.6}$$

So $\omega = \bigoplus_t \omega_t$ with $\omega_t \in \mathrm{Pos}(A_t E')$ and $\mathrm{tr}_{E'}\,\omega_t = \tau^{(t)}_A$; and the attack $\mathcal{E}$ and the attack $\mathcal{E}'$ that first measures the sector of the virtual signal reach the same state.

(iii) *(operators that are not reached)* the block diagonality of (ii) does not extend to operators merely constrained to have the marginal: in general a positive operator on AE' with marginal $\tau_A = \bigoplus_t \tau^{(t)}_A$ need not be block diagonal along $\mathbb{T}$, and this happens already for two one-dimensional sectors, $\tau_A = \frac{1}{2}\mathbb{1}_A$ and E' a qubit.

5 Assumptions

Assumption 5.1 [A-ENS] (Emission of the specified ensemble). For the declared signal ensemble $\{p_x, \rho_x\}$ of Definition 1.1 on the transmitted system, the letters the physical source draws and the systems it emits satisfy

the following; a letter that Alice obtains in untrusted-source mode by measuring a received system is not drawn by the source, and nothing below concerns it: (i) the letters $x_1, \ldots, x_n$ it draws in the n rounds are drawn independently, each with the weights p_x; (ii) given those letters, the systems it emits in the n rounds are in the state $\rho_{x_1} \otimes \cdots \otimes \rho_{x_n}$; (iii) neither the letter it draws in a round nor the state it emits in it depends on what happens in the other rounds, their outcomes and public messages included.

Assumption 5.2 [**A-SEQ**] (Sequential round structure)**.** The physical protocol has the sequential structure the declared protocol describes: (i) each round runs, in this order, state preparation and transmission; the measurements of the round, by every party that measures in it; the public messages of the round, in their declared order, *each of them sent after every measurement of the round is complete*, whichever party sends it and whichever party measured; and announcement-determined sifting; (ii) the classical post-processing is applied after the n rounds; (iii) every message is public; (iv) no operation on a quantum system of round j depends on a public message of round j; (v) every party that measures makes its measurements in round order: its measurement of round j is complete before its measurement of round $j + 1$ is.

Assumption 5.3 [**A-SRC**] (Retention of the system A in trusted-source mode)**.** For a source specification (Definition 1.1) in trusted-source mode, which fills the marginal slot with $\tau_A \in S_=(A)$, condition (SR3) of Definition 3.3 holds of the apparatus: (i) the system A is retained in the laboratory of the party whose label the party partition of the consuming declaration (Definition "Protocol declaration" (protocol-compiler/def:declaration)) gives A, and is never transmitted; (ii) no operation of the protocol or of the adversary acts on A except the declared measurement $\{M_x\}$ of that party.

Engine requirements

6 Certificate and licence requirements

Requirement 6.1 (Certificate and attestation demanded only at $d_A \geq 2$). The engine demands the source-replacement certificate of Definition 3.3 of a request in trusted-source mode only at $d_A \geq 2$, and there refuses such a request that carries none (`trusted_mode_without_certificate`). At $d_A = 1$, where the licensed condition $\omega_A = \tau_A$ excludes no state (Remark 4.7), neither the engine nor a consumer of its artefact demands the certificate or the attestation of Assumption 5.3.

Requirement 6.2 (The unique marginal served at $d_A = 1$). At $d_A = 1$ a request in trusted-source mode may leave the marginal slot empty. The engine then fills the slot with $\tau_A = [1]$, the unique element of $S_=(A)$ (Remark 4.7); the emitted specification is in trusted-source mode, trusted-source and untrusted-source mode imposing the same condition there; and the emitted artefact records that the slot was filled. This is the only slot the engine ever fills.

Requirement 6.3 (Refusal of a certificate that fails a decided condition). The engine refuses a request whose certificate is present and fails one of the conditions Definition 3.3 decides of its form — in pure form the three on Ψ, (SR1) and (SR2); in block form Hermiticity and positive semidefiniteness of every block, the trace, the block diagonality of every weighted virtual member along the matched sectors, and (3.1); in an ensemble presentation (E1) to (E3) — at every d_A, including $d_A = 1$, where no certificate is demanded (Requirement 6.1): a certificate that does not verify is a defect in the supplied data, not a weaker claim.

Requirement 6.4 (Direct decision of the declared marginal). Where τ_A is declared, the engine decides $\tau_A \in S_=(A)$ on the declared matrix directly, also where a certificate is present and satisfies (SR1), and never infers it from the implication (SR1) $\Rightarrow \tau_A \in S_=(A)$ of Theorem 4.1 (i); it refuses a request whose declared τ_A is not a state (`alice_marginal_not_admissible`). Consumers of the declared data decide it in the same way under their own rules; this requirement binds the engine.

Requirement 6.5 (Pass-through of the retention attestation). Where the engine emits the licence record `source_replacement_license`, which it does only in trusted-source mode at $d_A \geq 2$, the scope in which the certificate is demanded (Requirement 6.1), it records the (SR3) attestation it receives (Definition 3.3) in that record, and passes it through: it never verifies it and never asserts it of any physical apparatus (Assumption 5.3; Remark 4.6). No consumer can verify it either, since no function of the declared data decides it (Theorem 4.1 (i)).

Decoy-state sources and the tagging channel

7 The source model and the tagged declaration

Definition 7.1 (Phase-randomised weak-coherent-pulse source model)**.** A *weak-coherent-pulse model* is a source model, the case (b) of Definition 1.1

in which the specified ensemble $\{p_x, \rho_x\}$ is fixed as mathematical objects rather than as matrices, on a transmitted system whose Hilbert space is separable. It consists of the following declared data.

(W1) *(modes and polarisations)* two declared optical modes with a fixed orthonormal basis of the one-photon space $\mathbb{C}^2$, and a finite non-empty set Pol of *polarisations*, declared labels each standing for a unit vector $|a\rangle \in \mathbb{C}^2$, $a \in \mathrm{Pol}$, with creation operator $c_a^\dagger$. The vector $|a\rangle$ is a mathematical object of the model, as the states of (W4) are, and not an array of declared numbers: a declaration names the vector a label stands for and carries none of its coordinates, which need not be binary64 numbers ($(|H\rangle + |V\rangle)/\sqrt{2}$ has the coordinates $1/\sqrt{2}$). The transmitted system A' is the Fock space over the two modes, with vacuum $|\mathrm{vac}\rangle$; its Hilbert space is separable and infinite dimensional.

(W2) *(intensities and labels)* a finite non-empty set $\mathrm{Int} \subset [0, \infty)$ of *intensities*; the signal alphabet $\mathcal{X} \subseteq \mathrm{Pol} \times \mathrm{Int} \times \{\mathrm{test}, \mathrm{gen}\}$ with its declared order, an element of $\mathrm{Pol} \times \mathrm{Int} \times \{\mathrm{test}, \mathrm{gen}\}$ being a label and a label in $\mathcal{X}$ a letter, written $x = (a_x, \mu_x, z_x)$, whose third component is the *round type*; and weights $p_x > 0$ for every $x \in \mathcal{X}$ with $\sum_{x \in \mathcal{X}} p_x = 1$ in the declared order, rounding nowhere. A label of weight zero is not a letter of $\mathcal{X}$.

(W3) *(number states)* for $a \in \mathrm{Pol}$ and an integer $\mathsf{n} \geq 0$, the unit vector

$$|\mathsf{n}, a\rangle \;\coloneqq\; \frac{(c_a^\dagger)^{\mathsf{n}}}{\sqrt{\mathsf{n}!}}\, |\mathrm{vac}\rangle\,, \tag{7.1}$$

and the subspace $S_{\mathsf{n}} \coloneqq \mathrm{span}\{|\mathsf{n}, a\rangle : a \in \mathrm{Pol}\}$ of the n-photon subspace. Since the n-photon subspace of a two-mode Fock space has dimension $\mathsf{n} + 1$ and S_{n} is spanned by $|\mathrm{Pol}|$ vectors, $\dim S_{\mathsf{n}} \leq \min(\mathsf{n} + 1, |\mathrm{Pol}|)$; and $S_{\mathsf{n}} \perp S_{\mathsf{n}'}$ for $\mathsf{n} \neq \mathsf{n}'$, the two subspaces lying in different photon-number eigenspaces. $S_0 = \mathbb{C}\,|\mathrm{vac}\rangle$ is one dimensional, every polarisation sharing the vacuum.

(W4) *(the specified states)* with the *Poisson weight* $P_\mu(\mathsf{n}) \coloneqq e^{-\mu}\mu^{\mathsf{n}}/\mathsf{n}!$, the state prepared at the letter $x = (a, \mu, z)$ is

$$\rho_x \;=\; \sum_{\mathsf{n} \geq 0} P_\mu(\mathsf{n})\; |\mathsf{n}, a\rangle\langle \mathsf{n}, a|\,, \tag{7.2}$$

which does not depend on the round type z. It is a state: $\sum_{\mathsf{n} \geq 0} P_\mu(\mathsf{n}) = 1$ and each $|\mathsf{n}, a\rangle\langle \mathsf{n}, a|$ is a rank-one projector.

The physical-source assumption of the model, which asserts that the device emits (7.2), is Assumption 10.1; the model declares it in the place of Assumption 5.1.

Definition 7.2 (Tagged virtual source: cutoffs, flags and sector map)**.** Let a weak-coherent-pulse model be declared (Definition 7.1). A *tagged source declaration* over it consists of the following.

(T1) *(the two cutoffs)* integers N_{ent} and N_{src} with $0 \le N_{\mathrm{ent}} \le N_{\mathrm{src}}$: the *entropy cutoff*, the largest photon number whose sector stays quantum, and the *source cutoff*, the largest photon number that keeps a sector of its own.

(T2) *(the sector weights)* declared numbers $\tilde{p}(\mathsf{n} \mid \mu)$ for $0 \le \mathsf{n} \le N_{\mathrm{src}}$ and $\mu \in \mathrm{Int}$, depending on the intensity alone and not on the polarisation or the round type, satisfying

$$0 \;\le\; \tilde{p}(\mathsf{n} \mid \mu) \;\le\; P_\mu(\mathsf{n}) \qquad \text{for every } \mathsf{n} \le N_{\mathrm{src}},\ \mu \in \mathrm{Int}, \tag{7.3}$$

and the *tail weight*

$$\tilde{w}_\mu \;:=\; 1 - \sum_{\mathsf{n}=0}^{N_{\mathrm{src}}} \tilde{p}(\mathsf{n} \mid \mu), \tag{7.4}$$

the sum formed in increasing n and rounding nowhere. Then $\tilde{w}_\mu \ge 0$, because $\sum_{\mathsf{n} \le N_{\mathrm{src}}} \tilde{p}(\mathsf{n} \mid \mu) \le \sum_{\mathsf{n} \le N_{\mathrm{src}}} P_\mu(\mathsf{n}) \le 1$ by (7.3).

(T3) *(the sector index set and the virtual transmitted system)* a sector that no letter of $\mathcal{X}$ puts positive weight on is not declared: the *sector index set* carries the sectors the declaration uses and is read off the weights,

$$\begin{aligned} \mathbb{T} \;=\; & \{\mathsf{n} \le N_{\mathrm{ent}} :\ \tilde{p}(\mathsf{n} \mid \mu_x) > 0 \text{ for some } x \in \mathcal{X}\} \\ & \cup\, \{(\mathsf{n}, a) : N_{\mathrm{ent}} < \mathsf{n} \le N_{\mathrm{src}},\ \tilde{p}(\mathsf{n} \mid \mu_x) > 0 \text{ for some } x \in \mathcal{X} \text{ with } a_x = a\} \\ & \cup\, \{(a, \mu) : \tilde{w}_\mu > 0 \text{ and } (a, \mu) = (a_x, \mu_x) \text{ for some } x \in \mathcal{X}\}, \end{aligned} \tag{7.5}$$

in that declared order: the quantum sectors in increasing photon number, then the polarisation flags in the declared order of (n, a), then the tail flags in the declared order of (a, μ). The finite-dimensional virtual transmitted system is the orthogonal direct sum over it,

$$A'' \;=\; \bigoplus_{t \in \mathbb{T}} S_t, \qquad S_t = \begin{cases} S_{\mathsf{n}} & t = \mathsf{n} \le N_{\mathrm{ent}}, \\ \mathbb{C} f_{\mathsf{n},a} & t = (\mathsf{n}, a), \\ \mathbb{C} f_{a,\mu} & t = (a, \mu), \end{cases} \tag{7.6}$$

with the basis of each summand described next; it is the declared basis of A'' in the sense of Definition 3.3, the basis in which the blocks of the declaration are written. At a quantum sector $t = \mathsf{n}$ the summand is the space S_{n} of Definition 7.1 (W3), taken into the direct sum with its vectors $|\mathsf{n}, a\rangle$ and an orthonormal basis of S_{n}; like those vectors,

that basis is a mathematical object of the model and not an array of declared numbers (Proposition 8.1 returns S_{n} to the Fock space by an isometry V_{n}). There is one *polarisation flag* $f_{\mathsf{n},a}$ per photon number above the entropy cutoff and per polarisation of a letter that weights it, shared across intensities and round types; and one *tail flag* $f_{a,\mu}$ per pair a letter carries with $\tilde{w}_\mu > 0$, shared across round types. The flag vectors are declared orthonormal, the summands of (7.6) are mutually orthogonal in A'', and Π_t^{v} is the projector onto S_t. Every flag sector is one dimensional, and so is the quantum sector $\mathsf{n} = 0$. The index set is a function of the declared weights.

(T4) *(the virtual ensemble)* for $x = (a, \mu, z) \in \mathcal{X}$,

$$\rho_x^{\mathrm{v}} = \sum_{\mathsf{n} \le N_{\mathrm{ent}}} \tilde{p}(\mathsf{n} \mid \mu)\, |\mathsf{n}, a\rangle\langle \mathsf{n}, a| + \sum_{N_{\mathrm{ent}} < \mathsf{n} \le N_{\mathrm{src}}} \tilde{p}(\mathsf{n} \mid \mu)\, |f_{\mathsf{n},a}\rangle\langle f_{\mathsf{n},a}| + \tilde{w}_\mu\, |f_{a,\mu}\rangle\langle f_{a,\mu}|\,, \tag{7.7}$$

every sum running over the sectors of the index set (7.5) only: a quantum sector or a polarisation flag that no letter puts positive weight on is not a sector of A'' by (T3) and contributes no term, as the tail flag contributes none where $\tilde{w}_\mu = 0$. Each ρ_x^{v} is a state on A'' that is block diagonal along $\mathbb{T}$. Its component in a quantum sector $\mathsf{n} \le N_{\mathrm{ent}}$ is $\tilde{p}(\mathsf{n} \mid \mu)\, |\mathsf{n}, a\rangle\langle \mathsf{n}, a|$; in the polarisation flag sector (n, a') it is $\tilde{p}(\mathsf{n} \mid \mu)$ if $a' = a$ and zero otherwise; in the tail flag sector (a', μ') it is $\tilde{w}_\mu$ if $(a', \mu') = (a, \mu)$ and zero otherwise. It is positive semidefinite, being a sum of non-negative multiples of rank-one projectors, and its trace is $\sum_{\mathsf{n} \le N_{\mathrm{src}}} \tilde{p}(\mathsf{n} \mid \mu) + \tilde{w}_\mu = 1$ by (7.4), the absent terms contributing nothing to it: a sector that (7.5) leaves out is one on which $\tilde{p}(\mathsf{n} \mid \mu)$ vanishes for every letter, this one included, or a tail flag with $\tilde{w}_\mu = 0$. Two labels of the same polarisation and different intensities put weight on the same polarisation flag, and two labels of the same polarisation and intensity and different round types put weight on the same tail flag.

(T5) *(the matched sector map)* the retained system and its sector structure are not declared: a tagged source is presented as an ensemble presentation (Definition 3.3: a certificate defined from declared weighted blocks rather than declared), in which A, its structure, τ_A and $\{M_x\}$ are the objects (3.2) defines, A_t being a copy of S_t sector by sector and in the same order. In particular $\dim \mathcal{H}(A_t) = \dim \mathcal{H}(S_t)$ whatever the rank of the sector's marginal (Proposition 9.1 (iii)). The matched pair $(\{\Pi_t^A\}, \{\Pi_t^{\mathrm{v}}\})$, with Π_t^A the projectors onto Alice's sectors (Definition 1.1), is the structure of (T3) read on both sides. It is the retained-register sector map of the link of (T6), the slot (e) of Definition 2.1, which fixes how the certificate, τ_A and $\{M_x\}$ are presented, sector by sector. It is a statement about those three objects

only: it does not assert that any joint state of A and another system is block diagonal (Definition 1.1).

(T6) *(the link)* the channel Ξ_{tag} that Proposition 8.1 constructs and proves to carry each ρ_x^{v} to ρ_x, recorded as one source link (Definition 2.1) with the following slots: the kind tagging (slot (a)); the source descriptor $(A'', \{p_x, \rho_x^{\mathrm{v}}\})$ of (T3) and (T4) and the target descriptor $(A', \{p_x, \rho_x\})$ of the model (slot (b)); the channel witness supplied as a theorem constructor, namely Proposition 8.1, with an obligation record that names Assumption 10.1 (slot (c2)); the declared descriptor family, every tagged source declaration over every weak-coherent-pulse model (slot (d)); the retained-register sector map, the matched pair of (T5) (slot (e)); an empty list of transported constraint entries, so that no constraint is carried across the link, and hence no licence record, a licence being present only beside a non-empty list (slot (f)); and the status exact, the identity holding with no error term (slot (g)).

The cutoffs of (T1), the weights $\tilde{p}(\mathsf{n} \mid \mu)$ of (T2), and the model's alphabet, polarisation labels, intensities and label weights are *declared*. The polarisation vectors $|a\rangle$, the number states $|\mathsf{n}, a\rangle$ and the orthonormal basis of each quantum sector S_{n} of (T3) are objects of the model and are not declared numbers. The tail weight (7.4), the index set (7.5) and the weighted members $p_x \rho_x^{\mathrm{v}}$ of (7.7) are *formed*: the tail weight and the index set from the declared numbers in exact arithmetic, and each entry of a weighted member, in the basis of its sector, as the product of declared numbers with an entry of the dyad of the sector's signal vector in that basis, a number fixed by the model and the basis. They are the declaration's own data: read sector by sector, the weighted members are the blocks that the ensemble presentation of (T5) takes as its data. Everything on Alice's side — A, τ_A, $\{M_x\}$ and the certificate — is defined from those blocks and declared by no one.

8 The tagging channel

Proposition 8.1 (Tagging channel for weights at or below the Poisson weights). Let a weak-coherent-pulse model be given (Definition 7.1): labels $x = (a, \mu, z)$ with z the round type, unit polarisation vectors $|a\rangle$, Poisson weights $P_\mu(\mathsf{n}) = e^{-\mu}\mu^{\mathsf{n}}/\mathsf{n}!$ and model states $\rho_x = \sum_{\mathsf{n}\geq 0} P_\mu(\mathsf{n})\ |\mathsf{n}, a\rangle\langle \mathsf{n}, a|$ on the two-mode Fock space. Let a tagged source be declared on it (Definition 7.2) with cutoffs $0 \leq N_{\mathrm{ent}} \leq N_{\mathrm{src}}$, virtual states ρ_x^{v} of (7.7) and declared weights $\tilde{p}(\mathsf{n} \mid \mu)$, $\mathsf{n} \leq N_{\mathrm{src}}$, satisfying the inequality (7.3) of Definition 7.2 (T2), which reads

$$0 \leq \tilde{p}(\mathsf{n} \mid \mu) \leq P_\mu(\mathsf{n}) \qquad \text{for every intensity } \mu \text{ and every } \mathsf{n} \leq N_{\mathrm{src}}, \tag{8.1}$$

with tail weights $\tilde{w}_\mu = 1 - \sum_{\mathsf{n}\le N_{\mathrm{src}}} \tilde{p}(\mathsf{n}\,|\,\mu)$. Then:

(i) *(the tail states)* for every polarisation a and every intensity μ with $\tilde{w}_\mu > 0$,

$$\vartheta_{a,\mu} \coloneqq \frac{1}{\tilde{w}_\mu}\Big[\sum_{\mathsf{n}\le N_{\mathrm{src}}} \big(P_\mu(\mathsf{n}) - \tilde{p}(\mathsf{n}\,|\,\mu)\big)\,|\mathsf{n},a\rangle\langle\mathsf{n},a| + \sum_{\mathsf{n}>N_{\mathrm{src}}} P_\mu(\mathsf{n})\,|\mathsf{n},a\rangle\langle\mathsf{n},a|\Big] \tag{8.2}$$

is a state on the Fock space. Where $\tilde{w}_\mu = 0$ the declaration carries no tail flag for (a,μ) (Definition 7.2) and no tail state is defined.

(ii) *(one channel)* the map

$$\begin{aligned}\Xi_{\mathrm{tag}}(\zeta) \coloneqq &\sum_{\mathsf{n}\le N_{\mathrm{ent}}} V_\mathsf{n}\,\Pi^{\mathrm{v}}_\mathsf{n}\,\zeta\,\Pi^{\mathrm{v}}_\mathsf{n} V_\mathsf{n}^\dagger + \sum_{N_{\mathrm{ent}}<\mathsf{n}\le N_{\mathrm{src}}} \sum_a \langle f_{\mathsf{n},a}|\,\zeta\,|f_{\mathsf{n},a}\rangle\;|\mathsf{n},a\rangle\langle\mathsf{n},a| \\ &+ \sum_{(a,\mu):\,\tilde{w}_\mu>0} \langle f_{a,\mu}|\,\zeta\,|f_{a,\mu}\rangle\;\vartheta_{a,\mu},\end{aligned} \tag{8.3}$$

in which $\Pi^{\mathrm{v}}_\mathsf{n}$ projects $\mathcal{H}(A'')$ onto the quantum sector S_n, V_n is the isometry that embeds that sector into the Fock space as $\mathrm{span}\{|\mathsf{n},a\rangle\}_a$, and every sum runs over the sectors of the declared index set $\mathbb{T}$ only, is a channel from A'' to the Fock space, and

$$\Xi_{\mathrm{tag}}(\rho^{\mathrm{v}}_x) = \rho_x \qquad \text{for every label } x. \tag{8.4}$$

A quantum sector or a polarisation flag that no letter puts weight on is not a sector of A'' by Definition 7.2 (T3) and contributes no term to (8.3), exactly as for the tail flags. The channel Ξ_{tag} depends on the declared weights and the cutoffs and on no label.

(iii) *(the role of the inequality)* (8.4) holds for every family of weights satisfying (8.1). Where (8.1) fails at an intensity μ, its two halves fail differently.

(a) If $\tilde{p}(\mathsf{n}\,|\,\mu) > P_\mu(\mathsf{n})$ for some $\mathsf{n} \le N_{\mathrm{src}}$ and $\tilde{w}_\mu > 0$, the operator (8.2) is not positive semidefinite, for every polarisation a, and the construction is not available: where a letter x has intensity μ, the map (8.3) sends the state $|f_{a_x,\mu}\rangle\langle f_{a_x,\mu}|$ to $\vartheta_{a_x,\mu}$ and is not positive.

(b) If $\tilde{p}(\mathsf{n}\,|\,\mu) < 0$ for some $\mathsf{n} \le N_{\mathrm{src}}$ while $\tilde{p}(\mathsf{n}'\,|\,\mu) \le P_\mu(\mathsf{n}')$ for every $\mathsf{n}' \le N_{\mathrm{src}}$, then $\tilde{w}_\mu > 0$ and the operator (8.2) is a state for every polarisation a. What fails is the virtual state: for every letter x of intensity μ the operator ρ^{v}_x of (7.7) is not positive semidefinite or has trace greater than one, so it is not a state.

(iv) *(the published case)* for $\tilde{p}(\mathsf{n}\,|\,\mu) = P_\mu(\mathsf{n})$ and $N_{\mathrm{ent}} = N_{\mathrm{src}}$ the virtual states are those of [1, Eq. (145)] and (8.4) is [1, Lemma 9.2].

9 The certificate and the emitted declaration

Corollary 9.2 (The emitted decoy declaration). Let a weak-coherent-pulse model (Definition 7.1) and a tagged source declaration over it (Definition 7.2) be presented, with the certified inequality (7.3), $0 \le \tilde{p}(\mathsf{n} \mid \mu) \le P_\mu(\mathsf{n})$, discharged for every $\mathsf{n} \le N_{\mathrm{src}}$ and every intensity $\mu \in \mathrm{Int}$. Its *decoy declaration* is the ensemble presentation, in the sense of Definition 3.3, whose declared data are exactly

- the alphabet $\mathcal{X}$ with its declared order and the weights p_x;
- the matched sector structure, matched meaning that Alice's system and A'' carry one ordered index set: the sectors S_t of Definition 7.2 (T3) that some letter puts positive weight on, in the declared order of $\mathbb{T}$, each with its orthonormal basis of (T3);
- the weighted blocks $\sigma_x^{\mathrm{v},(t)} = p_x w_{x,t} \, |\varphi_{x,t}\rangle\langle\varphi_{x,t}|$ of (9.1), as block lists;
- the specified ensemble $\{p_x, \rho_x\}$ of the model as the source model, the source segment with the one tagging link of Definition 7.2 (T6) and its obligation record, and the assumption ledger;

so that no retained system, no measurement, no marginal and no certificate is among them. Below, *emitted* refers to this decoy declaration: the emitted source descriptor, segment and assumption ledger are the ones it declares. Then the following hold.

(i) *(a certificate, and an admissible specification)* where (E1) to (E3) of Definition 3.3 hold of the declared blocks, the objects (3.2) — the retained system $A = \bigoplus_t A_t$ with A_t a copy of S_t, the marginal τ_A, the POVM $\{M_x\}$ and the blocks $(\Psi_t)_t$ — are a source-replacement certificate in block form with every non-zero block a multiple of a rank-one projector, satisfying (SR1) and (SR2) exactly against the tagged virtual ensemble of Definition 7.2 (T4); the data are a source specification in the sense of Definition 1.1 in trusted-source mode; and Theorem 4.1 holds of it, its part (i) reading (E1) to (E3) as the decided conditions.

(ii) *(the segment, and the source identity at the emitted descriptors)*

(a) *(the source identity)* The emitted source descriptor is $(A'', \{\sigma_x^{\mathrm{v}}\})$ with $\sigma_x^{\mathrm{v}} = p_x \rho_x^{\mathrm{v}}$ for the tagged ensemble of Definition 7.2 (T4) with the declared weights, hence a member of the declared descriptor family of Proposition 8.1, every tagged source declaration over every weak-coherent-pulse model; so the source identity (2.3), here $\Xi_{\mathrm{tag}}(\rho_x^{\mathrm{v}}) = \rho_x$, holds exactly at the emitted descriptors, as that proposition's conclusion.

(b) *(the segment record)* The emitted segment satisfies the interface conditions (S1) (descriptor match), (S2) (index preservation) and (S4) (slots) of Definition 2.2, and its (S3) (source identity) is the conclusion of (a). Its list of transported constraint entries is empty and it carries no licence record. Its obligation record, the triple a consumer compares with the soundness registry to admit the link, is $(\mathsf{tagging},$ the id of Proposition 8.1, $\{\text{A-WCP}\})$, together with the member of the declared descriptor family that its descriptors are (Definition 2.1 (c2)). The third component is the union of the assumption closure of Proposition 8.1, which is empty, and the physical-source assumption A-WCP that this submodule declares for the model that proposition is stated over (Definition 2.2 (A1)). Inside an artefact that this submodule emits and whose assumption ledger lists A-WCP, the segment meets the admission conditions (A1) to (A3) of Definition 2.2: the registry entry, the decided triple and the provenance.

(iii) *(what the certificate reaches)* every state reached from the certificate of (i) by an attack on the transmitted system and the adversary's registers is block diagonal along $\mathbb{T}$ with A-marginal τ_A, and pinching A along $\mathbb{T}$ after the attack is the same operation as pinching the virtual signal before it (Lemma 4.5); Theorem 4.1 (iii) holds against the tagged ensemble of Definition 7.2 (T4) itself; and a positive operator constrained only to have the marginal τ_A need not be block diagonal.

(iv) *(the entries of the declared blocks)* (E1) to (E3) are conditions on the declared blocks, and every entry of a block is a product $p_x\, w_{x,t} \left(|\varphi_{x,t}\rangle\langle\varphi_{x,t}| \right)_{ij}$ of the declared weight p_x, the weight $w_{x,t}$, which is a declared sector weight or a tail weight formed from them, and an entry of the dyad of $|\varphi_{x,t}\rangle$ in the orthonormal basis of S_t of Definition 7.2 (T3), a number fixed by the model and that basis and not a declared number.

(a) *(the weights)* The products $p_x w_{x,t}$ are formed from the declared p_x, the declared $\tilde{p}(\mathsf{n} \mid \mu)$ and the tail weight (7.4), which is itself formed exactly and rounds nowhere. Choosing the p_x and the $\tilde{p}(\mathsf{n} \mid \mu)$ with few enough significant bits makes every product exact, and lowering a $\tilde{p}$ to a value that is still non-negative is permitted by (7.3) and yields again a tagged source declaration by (v).

(b) *(the signal dyads)* The entries of $|\varphi_{x,t}\rangle\langle\varphi_{x,t}|$ are the number 1 on every flag sector and on the vacuum, and on a quantum sector $\mathsf{n} \geq 1$ the entries of $|\mathsf{n}, a\rangle\langle\mathsf{n}, a|$ in the basis of S_{n}. For every unit vector $|b\rangle \in \mathbb{C}^2$, in Pol or not, write $|\mathsf{n}, b\rangle := (c_b^\dagger)^{\mathsf{n}} |\mathrm{vac}\rangle / \sqrt{\mathsf{n}!}$, which for $b \in \mathrm{Pol}$ is the vector of (7.1); then

$\big|\,\langle \mathsf{n}, a|\mathsf{n}, b\rangle\,\big|^2 = |\,\langle a|b\rangle\,|^{2\mathsf{n}}$ for all unit vectors a, b. Let every polarisation be one of the four vectors of $\{H, V, +, -\}$, with $|H\rangle, |V\rangle$ the fixed orthonormal basis of the one-photon space of Definition 7.1 (W1) and $|\pm\rangle = (|H\rangle \pm |V\rangle)/\sqrt{2}$, distinct polarisations being distinct vectors, and put $v_H = (1, 1, 0)$, $v_V = (1, -1, 0)$ and $v_\pm = (1, 0, \pm 1)$. If $|\mathrm{Pol}| \geq 2$, then S_1 is the one-photon space, $\{|1, H\rangle, |1, V\rangle\}$ is an orthonormal basis of it, and in that basis $|1, H\rangle\langle 1, H|$, $|1, V\rangle\langle 1, V|$ and $|1, \pm\rangle\langle 1, \pm|$ are $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $\frac{1}{2}\left(\begin{smallmatrix} 1 & \pm 1 \\ \pm 1 & 1 \end{smallmatrix}\right)$. If $|\mathrm{Pol}| \geq 3$, then S_2 is the two-photon space, and it has an orthonormal basis in which $|2, a\rangle\langle 2, a| = \frac{1}{2}\, v_a v_a^{\mathsf{T}}$ for every $a \in \{H, V, +, -\}$. So every entry of these dyads is 0, 1 or $\pm\frac{1}{2}$ at one photon and 0 or $\pm\frac{1}{2}$ at two.

(c) *(the sector sums)* With the entries of (a) and (b) representable, (E3) is a condition on the sums $\sum_x \sigma_x^{\mathrm{v},(t)}$, accumulated entry by entry in the declared order, and on nothing else.

(v) *(lowered weights)* if $0 \leq \tilde{p}'(\mathsf{n} \mid \mu) \leq \tilde{p}(\mathsf{n} \mid \mu)$ for every $\mathsf{n} \leq N_{\mathrm{src}}$ and every μ, then the weights $\tilde{p}'$ are again a tagged source declaration over the same model and Proposition 8.1 holds of it, the difference passing into the tail weight. Its weights, its virtual ensemble, its tail states and its sector list are those formed from $\tilde{p}'$: the sector list may gain a tail flag, where $\tilde{w}_\mu$ was zero and $\tilde{w}'_\mu$ is not, and it may lose a polarisation flag or a quantum sector, where a weight is lowered to zero at every letter that carried it, so that neither sector list need contain the other.

10 The physical-source assumption

Assumption 10.1 [**A-WCP**] (Emission of the declared weak-coherent-pulse states)**.** For the tagged source declaration of Definition 7.2, with the cutoffs, the weights and the polarisations it declares, the physical source satisfies: (i) in every round it emits the specified state ρ_x of Definition 7.1 of the letter $x = (a, \mu, z)$ drawn in that round, and nothing else; (ii) the intensity is the declared μ exactly, with no fluctuation and no uncertainty interval; (iii) the phase randomisation is full, so that the emitted state is the phase average (7.2); (iv) the emission is in the two optical modes declared in Definition 7.1 (W1), and the polarisation is the declared $|a\rangle$; (v) no degree of freedom other than those two modes carries any dependence on x; (vi) the round type z changes nothing the source emits; (vii) the letters $x_1, \ldots, x_n$ of the n rounds are drawn independently, each with the weights p_x of Definition 7.1 (W2); (viii) given those letters, the systems the source emits in the n rounds are in the product state $\rho_{x_1} \otimes \cdots \otimes \rho_{x_n}$ of the per-round states; (ix) neither the letter of a round nor the state emitted in it depends on what happens in the other rounds, their outcomes and public messages included.

11 Emission and refusals

Requirement 11.1 (Emission of the tagged ensemble as declared). When the engine emits an artefact from a tagged source declaration over a weak-coherent-pulse model, the artefact is that declaration's decoy declaration (Corollary 9.2) and nothing else: the engine declares no retained system, no measurement, no marginal and no certificate, and it rounds or otherwise alters no declared number. The matched sector structure among the declared data of that corollary is emitted as the retained-register sector map of Definition 7.2 (T5), with the block kind `proof_sector`, the tag that records a decomposition licensed by a proof, here source replacement, rather than by the device's own mode structure; it names Alice's sectors only as copies of the declared sectors S_t, so it declares no retained system. An artefact emitted from a tagged declaration carries A-WCP and not A-ENS in its assumption ledger (Assumption 10.1).

Requirement 11.2 (Order of the entropy and source cutoffs). The engine refuses, with no artefact, a declaration whose cutoffs are not integers with $0 \le N_{\mathrm{ent}} \le N_{\mathrm{src}}$, which is (T1) of Definition 7.2 (`cutoffs_out_of_order`, the one refusal for a cutoff that is not an integer and for cutoffs out of order).

Requirement 11.3 (Entropy cutoff of at most two photons). The engine refuses, with no artefact, a declaration with $N_{\mathrm{ent}} \ge 3$ (`entropy_cutoff_above_two`). The refusal is a scope decision and not a consequence of anything the claims state: every claim of this submodule is stated for arbitrary $N_{\mathrm{ent}} \le N_{\mathrm{src}}$ and nothing in them fails at $N_{\mathrm{ent}} \ge 3$. What is missing is a construction of the kind Corollary 9.2 (iv)(b) supplies for the four polarisations $\{H, V, +, -\}$ at one and at two photons: a declared basis in which the signal dyads have exactly declarable entries. This repository has no such construction at three photons and above. Admitting such a sector would mean deciding exact conditions on dyad entries that differ from the entries of the signal states, so until the construction exists the sector is refused rather than declared with entries that are not the numbers written.

Requirement 11.4 (Signal dyads formed by the exhibited construction). The engine forms the entries of the signal dyads of a decoy declaration itself, and takes neither a dyad nor a basis of a sector from a request. On the vacuum sector and on every flag sector it forms the number 1. On a quantum sector $\mathsf{n} \in \{1, 2\}$ of the index set it forms them only where every polarisation of Pol is declared as one of the four vectors of $\{H, V, +, -\}$, distinct polarisations as distinct vectors, and $|\mathrm{Pol}| \ge \mathsf{n} + 1$; it then forms the dyad of $|\mathsf{n}, a\rangle$ with the entries Corollary 9.2 (iv)(b) states for it. It decides of the formed dyads, exactly, that each is a rank-one projector on S_{n} (Hermitian,

idempotent and of unit trace) and that $\mathrm{tr}\left[\,|\mathsf{n},a\rangle\langle\mathsf{n},a|\;|\mathsf{n},b\rangle\langle\mathsf{n},b|\,\right] = |\,\langle a|b\rangle\,|^{2\mathsf{n}}$ for every pair of polarisations of Pol, the value that Definition 7.1 (W3) gives by that corollary and that is 0, $2^{-\mathsf{n}}$ or 1 for two vectors of $\{H, V, +, -\}$. It refuses, with no artefact, a declaration whose index set carries a quantum sector $\mathsf{n} \geq 1$ outside this construction, or at which one of these decisions fails (`quantum_sector_outside_construction`). A declaration with $N_{\mathrm{ent}} \geq 3$ is refused under its own reason (Requirement 11.3).

Requirement 11.5 (Certified bound on the sector weights). The obligation (7.3) of Definition 7.2 (T2), $0 \leq \tilde{p}(\mathsf{n} \mid \mu) \leq P_\mu(\mathsf{n})$ for every $\mathsf{n} \leq N_{\mathrm{src}}$ and every intensity $\mu \in$ Int of Definition 7.1 (W2), is an inequality and never an equality: its right-hand side is not a declared datum, and at a positive intensity, declared as a binary64 number, it is a transcendental number (at $\mu = 0$ it is 1 for $\mathsf{n} = 0$ and 0 for $\mathsf{n} \geq 1$). For every such (n, μ) the engine forms, from the declared intensity μ and in exact rational arithmetic, the lower end $\underline{P}_\mu(\mathsf{n}) := \max(0, S_{K+1})\,\mu^{\mathsf{n}}/\mathsf{n}!$ of the enclosure of Lemma 7.3, with S_{K+1} the partial sum (7.8) at $K = K_\mu$, the least even integer $K \geq \mu$ with $t_{K+1} = \mu^{K+1}/(K+1)! \leq 2^{-90}$, which exists because $t_{K+1} \to 0$. It records K_μ for every intensity in the artefact, and it reads no enclosure of $P_\mu(\mathsf{n})$ from the request. By part (iv) of that lemma no even K with $\mu \leq K < K_\mu$ gives a larger lower end, so the rule refuses no weight that a smaller admissible index would admit. It refuses, with no artefact, a declaration in which $\tilde{p}(\mathsf{n} \mid \mu) < 0$ or $\tilde{p}(\mathsf{n} \mid \mu) > \underline{P}_\mu(\mathsf{n})$ at some (n, μ) (`sector_weight_above_poisson`). Since $\underline{P}_\mu(\mathsf{n}) \leq P_\mu(\mathsf{n})$ by that lemma, every declaration violating (7.3) is refused, and every declaration it admits satisfies (7.3). It refuses, with no artefact, a declaration presenting the obligation as an equality $\tilde{p}(\mathsf{n} \mid \mu) = P_\mu(\mathsf{n})$ to be decided (`sector_weight_as_equality`). No condition of this submodule compares $\tilde{p}(\mathsf{n} \mid \mu)$ with $P_\mu(\mathsf{n})$ for equality and none is decided at a tolerance.

Requirement 11.6 (Shared polarisation and tail flags). The engine refuses, with no artefact, a declaration carrying a flag family finer than the one Definition 7.2 (T3) declares (`flag_family_split_by_intensity`). That family is one polarisation flag $f_{\mathsf{n},a}$ for each sector (n, a) of the index set of (T3), $N_{\mathrm{ent}} < \mathsf{n} \leq N_{\mathrm{src}}$, shared by every intensity and both round types, and one tail flag $f_{a,\mu}$ for each sector (a, μ) of that index set, shared by both round types; a finer family is one with separate flags where (T3) declares one. In particular the engine refuses a family whose polarisation flags are split by intensity at a photon number $N_{\mathrm{ent}} < \mathsf{n} \leq N_{\mathrm{src}}$. A family split by intensity still satisfies the source identity but gives up the relation across intensities that is the decoy content of the declaration, one photon number's weight at every intensity sitting on one flag (Remark 8.2), and without the refusal that loss would be silent.

Requirement 11.7 (Binary64 entries of the emitted blocks and sector sums). The engine decides (E3) of Definition 3.3 on the sums of the declared blocks of a decoy declaration, accumulated entry by entry in the declared order. Each entry of a block, a product of declared numbers, the tail weight formed from them and a formed entry of a signal dyad (Corollary 9.2 (iv)), is required to be a binary64 number and is refused, with no artefact, when it is not (`formed_entry_not_declarable`). Those sums are required to round nowhere, so that their entries are binary64 numbers, and a declaration at a sector of which the accumulation rounds fails (E3) and is refused, with no artefact (`ensemble_presentation_not_admissible`). No entry is ever rounded, because rounding an entry of a declared block would change which ensemble is declared and so which ensemble the source identity of Corollary 9.2 (ii) is asserted of.

Requirement 11.8 (Positivity and weight sums of the emitted blocks). The engine decides (E1) and (E2) of Definition 3.3 on the declared blocks of a decoy declaration, as it decides (E3) (Requirement 11.7). It refuses, with no artefact, a declared block failing (E1) or a weight sum failing (E2) (`ensemble_presentation_not_admissible`).

References

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[2] Adam Winick, Norbert Lütkenhaus, and Patrick J. Coles. Reliable numerical key rates for quantum key distribution. *Quantum 2, 77 (2018)*, 2018.